

Tuyet Minh Vu

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EDUCATION

University of Utah – John & Marcia Price College of Engineering

Bachelor of Science in Computer Science

Salt Lake City, UT

Expected May 2028

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Software Practice, Machine Learning

TECHNICAL SKILLS

Languages: Python, Java, C#, C++, JavaScript, HTML/CSS

Frameworks & Libraries: PyTorch, NumPy, Scikit-learn, Blazor, React

Tools & Technologies: Git/GitHub, GitLab, Unity, SQL, Figma, Qt

EXPERIENCE

University of Utah – Therapeutic Games and Apps Lab

Undergraduate Software Developer / Research Assistant

Salt Lake City, UT

May 2026 – Present

- Contributed to *Color Me Healthy*, a Unity/C# pediatric health game enabling children ages 5–10 to self-report symptoms through interactive check-ins, deployed in active clinical use.
- Implemented features across the full application stack in C# and Unity, including symptom flows, notification panels, localization logic, and WebGL configuration.
- Identified, reproduced, and documented bugs across drawing, persistence, and account systems, accelerating resolution cycles between developers and clinical partners.
- Managed work through Git/GitLab with feature branches, build testing, and code review in an agile research environment.

Salt Lake Community College – Programming Club

Officer

Salt Lake City, UT

September 2024 – May 2025

- Designed and led weekly LeetCode-style coding challenges in data structures and algorithms for 200+ students, improving technical interview readiness across the club.
- Authored problem sets with edge cases, test suites, and solution walkthroughs covering time and space complexity to build rigorous algorithmic thinking.
- Grew club engagement through targeted social media campaigns, classroom outreach, and coordination of coding competitions and technical workshops.

PROJECTS

Learning Management System | C#, SQL, Blazor

April 2026

- Architected a normalized relational database with constraints and indexing to enforce data integrity and support scalable multi-table queries across courses, users, and assignments.
- Implemented automated grading logic to dynamically compute student scores from submissions, assignment categories, and configurable weights, eliminating manual calculation.
- Integrated a Blazor front-end with SQL backend to surface real-time grade updates, enrollment data, and course dashboards with clean component-level data binding.

Cooking Educational Game | C++, Qt, Box2D

April 2026

- Built a physics-based cooking education game in C++ with Qt and Box2D that teaches recipe concepts through real-time ingredient simulations, delivering a polished end-to-end experience.
- Engineered live physics using Box2D forces and frame-by-frame motion updates to simulate pouring, mixing, and whisking with distinct particle effects for dry and liquid ingredients.
- Designed a custom cursor interaction system with timer-based animations for ingredient selection, increasing engagement through tactile, responsive feedback.

Real-Time Multiplayer Snake Game | C#, Blazor, SQL, TCP Networking

November 2025

- Architected a real-time multiplayer client in C# using asynchronous TCP networking and an event-driven model to handle concurrent player connections with minimal latency.
- Built a dynamic Blazor UI rendering snakes, walls, and power-ups in real time with low-latency keyboard input keeping player state synchronized with the server.
- Optimized rendering by deserializing JSON server broadcasts and triggering targeted component redraws, reducing unnecessary re-renders and improving frame consistency.
- Integrated a SQL backend via REST endpoints to persist per-game and per-player statistics, powering a live leaderboard and post-game analytics dashboard.